Physics-Informed (and Residual) Diffusion for OD-Conditioned Trajectory Generation

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Roadmap (Phases)

- **Phase A (pipeline)**: build a leakage-free, dt-fixed protocol and sanity-check data products.
- **Phase B v1.0 (models + diagnosis)**: baseline vs diffusion vs physics diffusion; identify the shrinkage failure mode.
- **Phase B v1.1 (fix)**: introduce residual diffusion (frozen prior + stochastic residual) to recover macroscopic realism.

Task Definition

- Task: **KnownDestination** OD-conditioned trajectory generation (grid space).
- Windows: history H=8, horizon F=12 (dt-fixed $\Delta t = 30$s).
- Goals:
  - **Coverage**: best-of-K (oracle) ADE/FDE, distribution metrics (Fréchet/DTW).
  - **Physical realism**: macroscopic motion (MSD curve, Rog).

Outline

Phase A: Pipeline

Phase B v1.0: Models & Diagnosis

Phase B v1.1: Residual Diffusion Fix

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Phase A: Pipeline

Phase B v1.0: Models & Diagnosis

Phase B v1.1: Residual Diffusion Fix

Phase A (from raw data to strict pipeline)

Relation to previous stage: starting from raw Shenzhen taxi GPS records, we need a reproducible pipeline before any model comparison.

Phase A objective: eliminate ambiguity and leakage (dt, split, nav_field source, normalization).

Phase A outputs:

- dt-fixed dataset ($\Delta t = 30$s) and train/val/test splits
- train-only products: data_stats.json (normalizer) + nav_field.npz
- sanity checks (split overlap, metadata source, coordinate range, alignment)

Phase A: Strict Data Contract (KISS)

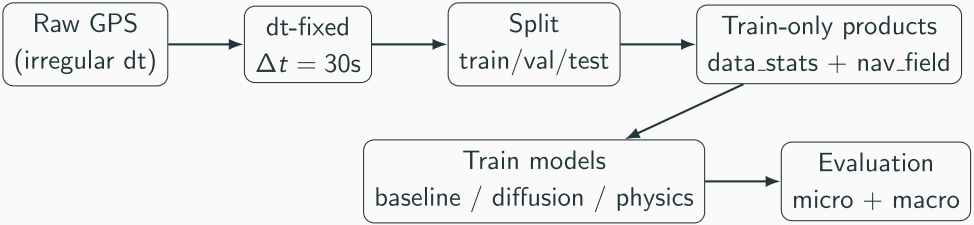


Figure 1: *

Split-first, train-only products: prevents nav_field/statistics leakage into val/test.

Outline

Phase B v1.0 (what changes from Phase A?)

Relation to Phase A: with a strict dt-fixed pipeline, Phase B compares model families under the same protocol.

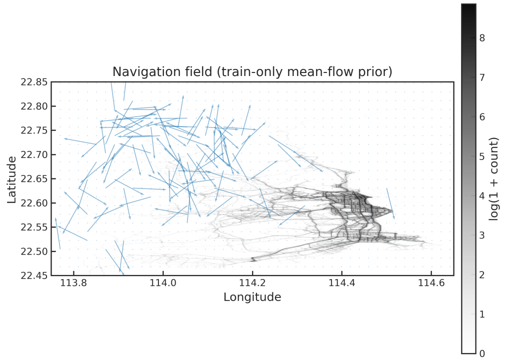
Phase B v1.0 question: does diffusion provide better multi-modal coverage, and does nav_field improve it?

Models compared:

- Baseline (deterministic L2 regression, $K{=}1$)
- Diffusion (data-only, $K{=}20$)
- Physics diffusion (nav_field conditioning, $K{=}20$)

Phase B v1.0: What is “Physics” here? (Nav field prior)

- **Nav field** is estimated on the **train split only**: local mean-flow direction + count/speed statistics.
- At inference we crop a local patch around the last observed position and encode it as conditioning.
- Interpretation: a **local mean-field prior** that stabilizes directionality and improves best-of-K coverage.



Phase B v1.0: Micro-Level Performance (coverage)

- Deterministic baseline has strong mean ADE/FDE (as expected for L2 regression).
- Diffusion/physics diffusion improve **best-of-K** (oracle) capability.
- Physics improves oracle upper bound over data-only diffusion.

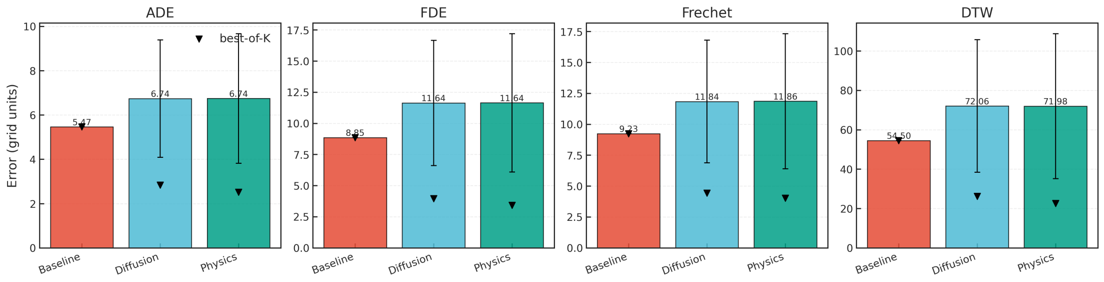


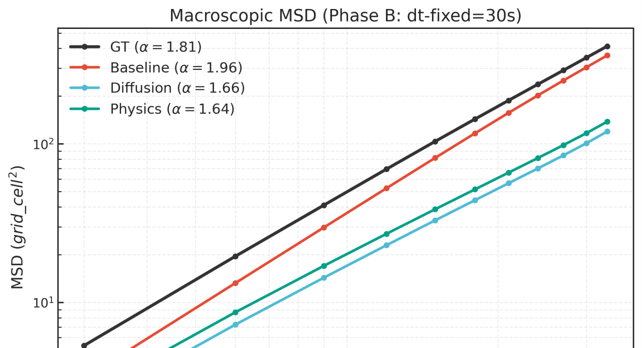
Figure 2: *

Micro-level metrics (mean/std and best-of-K where applicable).

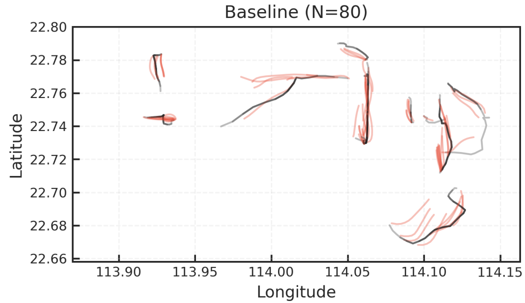
Phase B v1.0: Macro-Level Failure (shrinkage)

Key finding: despite strong best-of-K, diffusion samples under-estimate displacement (low MSD/Rog).

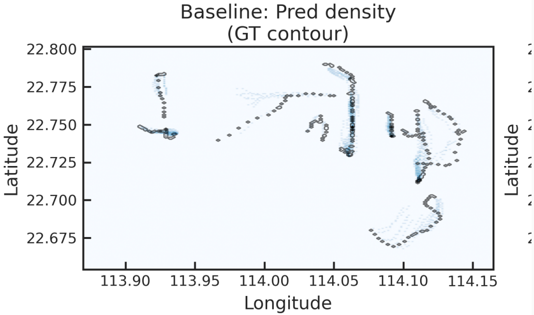
- This is a **systematic** low-frequency bias (not a single-case artifact).
- It threatens the "physics" story: coverage without physical realism.



Phase B v1.0: Interpretable Maps (Baseline)



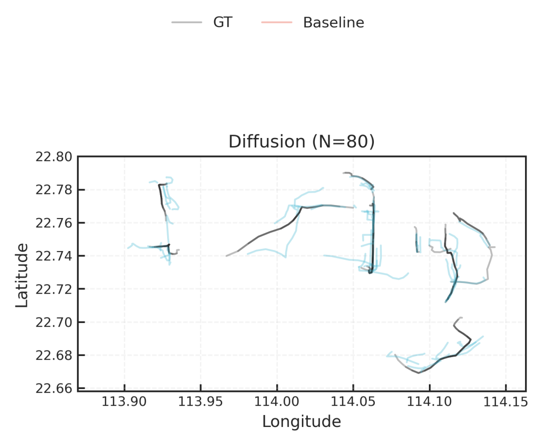
Trajectory overlay (Baseline vs GT)



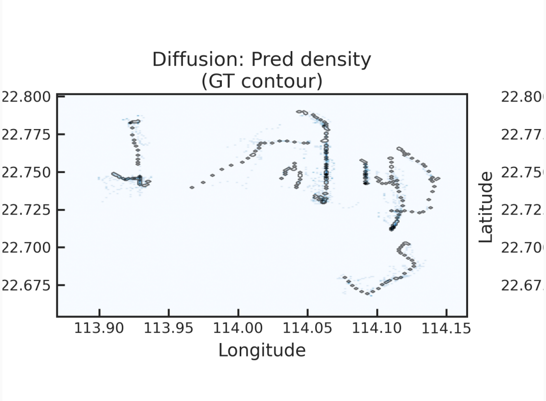
Density (Pred heatmap, GT contour)

Reading guide: these are "map-like" visualizations via bbox projection (no road map-matching). They highlight spatial heterogeneity and corridor structure.

Phase B v1.0: Interpretable Maps (Diffusion)



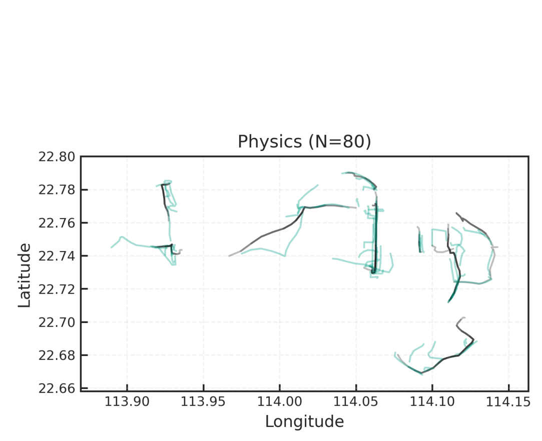
Trajectory overlay (Diffusion vs GT)



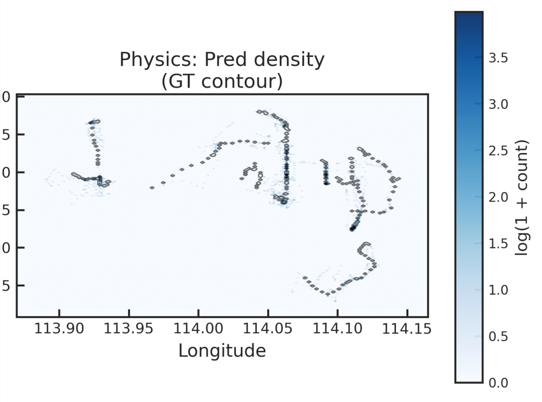
Density (Pred heatmap, GT contour)

Reading guide: diffusion produces diverse futures, but macro displacement can be biased low (shrinkage). *(Middle panel of combined figure.)*

Phase B v1.0: Interpretable Maps (Physics)



Trajectory overlay (Physics vs GT)



Density (Pred heatmap, GT contour)

Reading guide: nav_field acts as a local mean-flow prior and improves oracle coverage, but does not eliminate shrinkage. *(Right panel of combined figure.)*

Phase B v1.0: Qualitative Cases (grid space)

Each panel is a different OD-conditioned sample (GT vs Baseline vs Diffusion vs Physics). This helps visually connect micro-level coverage to the macro-level shrinkage diagnosis.

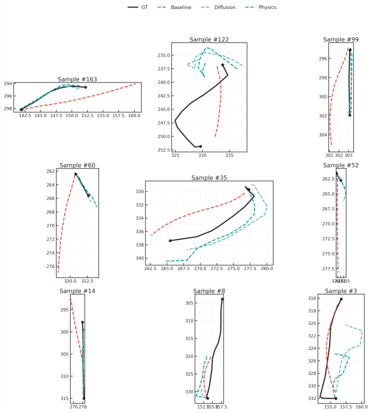
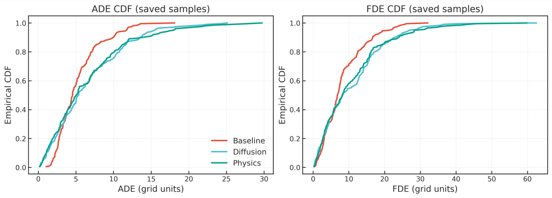
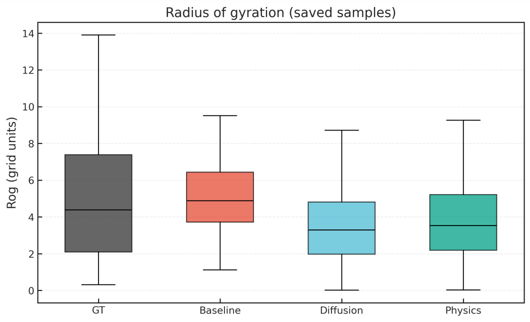


Figure 4: *

Phase B v1.0: Distribution Diagnostics



Error CDF (saved samples)



Rog distribution (saved samples)

Phase B v1.0: Why macro-loss ablation is insufficient

- Adding a macro loss on x_0^{pred} is conceptually valid, but **noisy** at large timesteps ($t \approx T$).
- If not carefully gated/normalized, the model can satisfy macro loss with **high-frequency shortcuts** (jitter).
- Empirically: limited improvements (plateau region), motivating an architectural pivot.

Outline

Phase B v1.1 (link from v1.0 to fix)

Relation to Phase B v1.0: v1.0 reveals a low-frequency bias (shrinkage) that is hard to correct via noisy macro proxies.

Phase B v1.1 objective: recover macroscopic displacement *without* sacrificing diversity and best-of-K coverage.

Key idea: let a deterministic model handle low-frequency scale; let diffusion model only stochastic residuals.

Residual Diffusion (v1.1): Method

Decomposition

$$\mathbf{v}_{1:F} = \mathbf{v}^{\text{prior}}_{1:F} + \mathbf{v}^{\text{res}}_{1:F}, \qquad \mathbf{v}^{\text{prior}}_{1:F} = f_{\text{base}}(\mathbf{X}, \mathbf{c})$$

- Freeze f_{base} (baseline) as a deterministic prior (no gradient leakage).
- Train diffusion on residual targets: $\mathbf{v}^{\text{res}} = \mathbf{v} - \mathbf{v}^{\text{prior}}$.
- Inference: sample $\mathbf{v}^{\text{res}}$ then add back $\mathbf{v}^{\text{prior}}$.

Residual Diffusion (v1.1): Quick Validation

Model	K	ADE_{mean}	ADE_{best}	FDE_{mean}	FDE_{best}	MSD_{10}	Rog
Baseline (prior, L2)	1	4.602	4.602	7.239	7.239	507.712	5.345
Residual Diffusion	20	5.735	2.496	9.680	3.053	576.014	6.225

- Same conditions (val subset, 192 samples). Residual keeps strong best-of-K.
- Macro recovery: Rog/GT $\approx$ 0.93, MSD_{10}/GT $\approx$ 0.84, speed ratio $\approx$ 0.95.
- Diversity check: ADE_{std} is non-zero (no residual collapse observed in quick eval).

Phase B v1.1: Interpretable Maps (Residual Diffusion) — Overlay

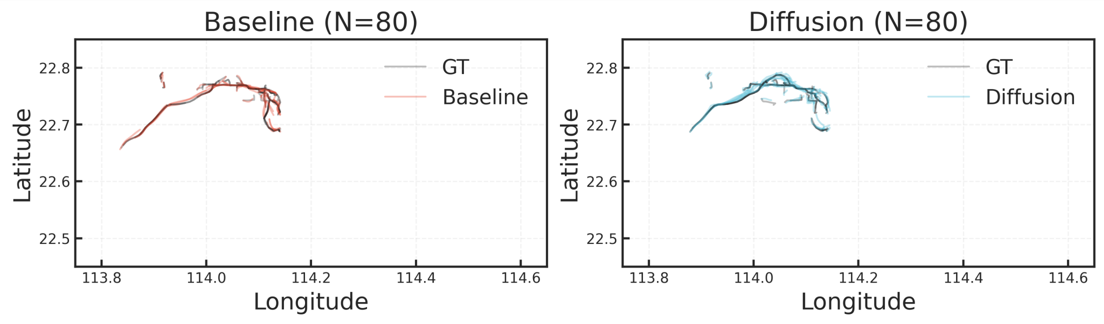


Figure 5: *

Left: Baseline (prior). Right: Residual Diffusion (v1.1). **Note:** “Diffusion” label in figure refers to residual diffusion.

Phase B v1.1: Interpretable Maps (Residual Diffusion) — Density

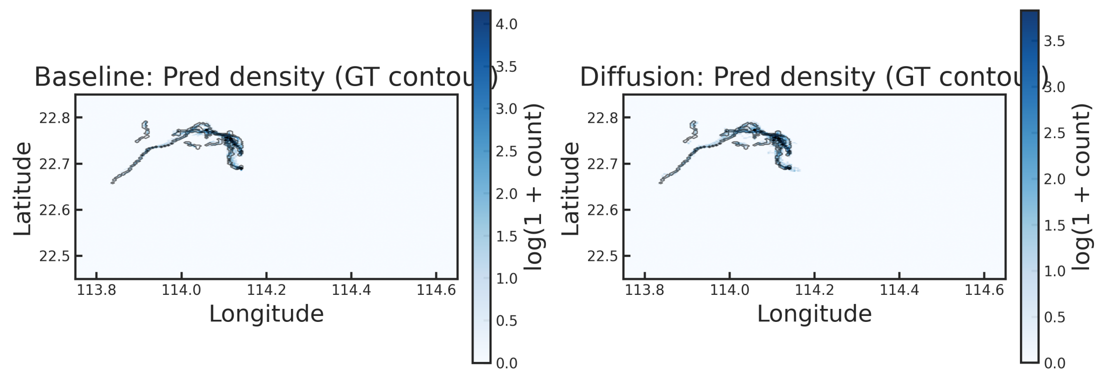


Figure 6: *

Left: Baseline (prior). Right: Residual Diffusion (v1.1). Pred density heatmap with GT contour.

Residual Diffusion: What we learned

- The shrinkage is fundamentally a **low-frequency learning bias** (mean reversion).
- Residual decomposition is a **principled fix**: it separates "scale" (prior) from "uncertainty" (residual).
- Physics/nav_field remains useful as a **directional guide** once scale is stabilized.

Takeaways & Next Steps

- Phase A: strict, leakage-free dt-fixed pipeline enables fair comparisons.
- Phase B v1.0: diffusion improves best-of-K but fails macro realism (shrinkage).
- Phase B v1.1: residual diffusion recovers macro scale while preserving diversity.
- Next steps (paper-ready):
  - Residual **physics diffusion** (nav_field + residual prior).
  - Strong generative baselines (SeqCVAE; recent diffusion predictors).
  - Robustness: prior-swap test and additional geographic visualizations (zoomed regions / map tiles).

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